

Learning DevOps can feel overwhelming, but with a structured plan, it becomes manageable. Below is a comprehensive, step-by-step **6-month guide** that covers everything from project management to monitoring, including all the tools we discussed.

This schedule is designed for a commitment of **15–20 hours per week**. Adjust the pace to fit your availability—it can be stretched to 8–9 months if you have less time.

 The 6-Month DevOps Learning Schedule

Phase	Month	Week	Topic	Key Tools & Technologies	Hands-On Project / Task
1. Foundations	1	1	DevOps Culture & Project Management	Agile, Scrum, Kanban, Jira, Trello, Confluence	Set up a Jira project board. Create user stories and a sprint backlog for a sample application.
		2	Linux Fundamentals & Networking	Linux CLI (Ubuntu/CentOS), Bash, SSH, Permissions, Systemd	Navigate the file system, manage users, set up a web server (Nginx/Apache) manually.
		3	Version Control with Git	Git, GitHub, GitLab, Branching (Git Flow)	Create a repo, commit changes, create branches, merge, and handle merge conflicts.
		4	Scripting & Automation	Bash Scripting, Python Basics	Write a Bash script to automate system backups. Write a Python script to parse log files.
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2. CI/CD & Build	2	5	Build Systems & Package Management	C++: CMake, Conan Python: Poetry, pip JavaScript: Vite, npm	Set up a build system for a sample project in your chosen language. Manage dependencies.
		6	Testing Fundamentals	C++: Google Test Python: pytest JavaScript: Jest	Write unit tests and integration tests for your sample project. Aim for >80% code coverage.
		7	Continuous Integration (CI)	Jenkins, GitHub Actions, GitLab CI	Create a CI pipeline that automatically builds and tests your code on every push.

		8	Continuous Delivery (CD)	GitHub Actions, GitLab CI, ArgoCD, Spinnaker	Extend your pipeline to automatically deploy your application to a staging environment.
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3. Containers & Orchestration	3	9	Containerization with Docker	Docker, Docker Compose, Container Registries	Containerize your sample application. Use Docker Compose to run multi-container apps.
		10	Container Orchestration (Intro)	Kubernetes (K8s), Minikube, kubectl	Deploy a containerized app to a local Kubernetes cluster. Learn Pods, Deployments, Services.
		11	Advanced Kubernetes	Helm, Ingress, Persistent Volumes, ConfigMaps	Package your app with Helm. Set up ingress for external access. Manage configuration.
		12	Kubernetes in Production	K3s, EKS, AKS, GKE	Deploy a scalable, highly available application on a managed Kubernetes service (e.g., EKS).
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4. Infrastructure as Code	4	13	Configuration Management	Ansible, Chef, Puppet	Use Ansible to automate the setup of a web server (install Nginx, configure firewall).
		14	Infrastructure as Code (IaC)	Terraform, Pulumi, AWS CloudFormation	Provision a cloud environment (VPC, subnets, EC2 instances) using Terraform.
		15	Cloud Platforms (AWS/Azure/GCP)	AWS (EC2, S3, RDS, VPC), Azure, GCP	Deploy a sample application to the cloud. Use managed services like RDS for a database.
		16	GitOps	ArgoCD, Flux	Implement GitOps: use Git as the single source of truth for your infrastructure and application state.

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5. Monitoring & Debugging	5	17	Logging	ELK Stack (Elasticsearch, Logstash, Kibana), Loki	Centralize logs from your application and infrastructure. Create dashboards in Kibana.
		18	Metrics & Monitoring	Prometheus, Grafana	Set up Prometheus to collect metrics (CPU, memory, request latency). Visualize them in Grafana.
		19	Alerting & Incident Management	Alertmanager, PagerDuty, Opsgenie	Configure alerts for high error rates or system downtime. Integrate with PagerDuty for on-call rotations.
		20	Debugging & Distributed Tracing	Jaeger, Zipkin, Sentry, GDB/Valgrind (C++)	Implement distributed tracing to track requests across microservices. Use Sentry for error tracking.
Phase	Month	Week	Topic	Key Tools & Technologies	Hands-On Project / Task
6. Advanced & Integration	6	21	Security (DevSecOps)	SonarQube, Snyk, Trivy, OWASP ZAP	Integrate security scanning into your CI/CD pipeline (SAST, DAST, dependency scanning).
		22	AI & MLOps (Optional)	MLflow, Kubeflow, NVIDIA Triton, DVC	Learn how DevOps principles apply to machine learning. Set up a model training and serving pipeline.
		23	Final Project (Part 1)	All tools combined	Project: Design and implement a complete end-to-end CI/CD pipeline for a microservices application on Kubernetes with monitoring and alerting.
		24	Final Project (Part 2)	All tools combined	Project: Polish the project, document everything, and

					present it as a portfolio piece.
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The Recommended Learning Path

This schedule follows a proven, logical progression:

1. **Start with the Fundamentals:** Linux, networking, and a scripting language like **Bash** or **Python** are your foundation. Without these, the rest will be very difficult.
2. **Master Version Control:** Get comfortable with **Git** and a platform like **GitHub** or **GitLab**. This is non-negotiable.
3. **Learn Build & Test:** Understand how to compile code and run tests automatically. This prepares you for CI/CD.
4. **Build CI/CD Pipelines:** Use a tool like **GitHub Actions** or **Jenkins** to automate the build, test, and deployment process.
5. **Learn Containerization:** Begin with **Docker** to understand how to package applications.
6. **Move to Orchestration:** Once comfortable with Docker, learn **Kubernetes** to manage containers at scale.
7. **Tackle Infrastructure as Code:** Learn **Terraform** to provision the infrastructure your containers will run on.
8. **Implement Monitoring & Logging:** Finally, learn **Prometheus** and **Grafana** to observe your running applications.

Tips for Success

- **Hands-on > Theory:** DevOps is a practical field. Spend at least 70% of your time doing hands-on labs and projects.
- **One Tool at a Time:** Don't try to learn everything at once. Master one tool before moving to the next.
- **Use Free Resources:**
 - **Roadmap.sh:** The [DevOps Roadmap](#) is an excellent visual guide.
 - **GitHub:** Explore the many free DevOps learning paths and resources available.
 - **Cloud Free Tiers:** AWS, Azure, and GCP all offer free tiers perfect for learning.
- **Build a Portfolio:** The final project is crucial. It demonstrates your ability to apply all the tools in a real-world scenario.

- **Join a Community:** Engage with DevOps communities on Reddit (r/devops), LinkedIn, or Discord. Ask questions and learn from others.

What About My Programming Language?

You can apply this roadmap with **any** language (C++, Python, JavaScript, etc.). The tools for **build, test, and package management** will change, but the CI/CD, containerization, and cloud concepts remain the same. Choose one language to focus on for your projects.

This schedule provides a clear, step-by-step path. Would you like me to elaborate on any specific week or tool?

Below is a comprehensive, end-to-end DevOps tools table, classified from **Project Management** to **Monitoring**. For each category, I've highlighted the **most popular** tools specifically for **C++**, **Python**, **Web (JavaScript)**, and **AI/ML** development in 2025.

 The Complete DevOps Toolchain (Classified by Stage)

Stage	Category	C++	Python	JS/TypeScript	AI / ML
Plan	Project Management	Jira, Linear, Azure DevOps	Jira, Linear, Azure DevOps	Jira, Linear, Azure DevOps	Jira, Linear, Azure DevOps
Plan	Documentation & Knowledge	Confluence, Notion, GitHub Wiki	Confluence, Notion, GitHub Wiki	Confluence, Notion, GitHub Wiki	Confluence, Notion, GitHub Wiki
Code	Version Control (SCM)	Git, GitHub, GitLab, Bitbucket	Git, GitHub, GitLab, Bitbucket	Git, GitHub, GitLab, Bitbucket	Git, GitHub, GitLab, Bitbucket
Code	IDE & Local Environment	Visual Studio, CLion, VS Code	PyCharm, VS Code, Jupyter	VS Code, WebStorm, Cursor	VS Code, Jupyter, Cursor
Code	AI Coding Assistant	GitHub Copilot, SourceCraft	GitHub Copilot, Amazon Q	GitHub Copilot, Cursor AI	GitHub Copilot, Cursor AI, Sourcegraph Cody
Build	Package Manager	Conan, vcpkg, Build2	pip, Poetry, Conda	npm, yarn, pnpm	pip, Conda, Poetry
Build	Build System / Compiler	CMake, Ninja, Bazel, Make	Setuptools, Poetry, PyInstaller	Vite, Webpack, esbuild, Rollup	Setuptools, Poetry, Bazel
Build	Code Quality & Linting	Clang-Tidy, Clang-Format, SonarQube	Ruff, Flake8, Black, mypy	ESLint, Prettier, SonarQube	Ruff, Pylint, mypy
Test	Unit / Integration Testing	Google Test, Catch2, CTest	pytest, unittest, nose	Jest, Vitest, Mocha	pytest, unittest
Test	E2E / Browser Automation	Selenium, Playwright	Selenium, Playwright	Playwright, Cypress, Selenium	Selenium, Playwright
Test	Performance / Load Testing	JMeter, Gatling, Valgrind	JMeter, Locust	JMeter, k6	JMeter, k6
Test	Security / SAST	SonarQube, CodeQL	Bandit, SonarQube, CodeQL	SonarQube, CodeQL, Snyk	Bandit, SonarQube, CodeQL
CI	Continuous Integration	GitHub Actions, GitLab CI, Jenkins	GitHub Actions, GitLab CI, Jenkins	GitHub Actions, GitLab CI, CircleCI	GitHub Actions, GitLab CI, Jenkins

CD	Continuous Delivery / GitOps	ArgoCD, Flux	ArgoCD, Flux, Spinnaker	ArgoCD, Flux, Vercel, Netlify	ArgoCD, Flux, Kubeflow
Deploy	Containerization	Docker, Podman	Docker, Podman	Docker, Podman	Docker, Podman, NVIDIA Container Toolkit
Deploy	Container Orchestration	Kubernetes, K3s	Kubernetes, K3s	Kubernetes, K3s	Kubernetes, K3s
Deploy	Infrastructure as Code (IaC)	Terraform, Pulumi	Terraform, Pulumi, AWS CDK	Terraform, Pulumi	Terraform, Pulumi
Deploy	Configuration Management	Ansible, Chef	Ansible, SaltStack, Fabric	Ansible	Ansible
Deploy	ML Model Serving	—	—	—	NVIDIA Triton, TensorFlow Serving, BentoML
Deploy	ML Pipeline Orchestration	—	—	—	Apache Airflow, Kubeflow, MLflow, Metaflow
Operate	Debugging & Tracing	GDB, Valgrind, AddressSanitizer	pdb, py-spy, Sentry	Chrome DevTools, Sentry	LangSmith, Weights & Biases
Operate	Distributed Tracing	Jaeger, Zipkin	Jaeger, Zipkin	Jaeger, Zipkin	Jaeger, Zipkin
Monitor	Metrics & Monitoring	Prometheus + Grafana	Prometheus + Grafana	Prometheus + Grafana	Prometheus + Grafana
Monitor	Logging	ELK Stack (Elasticsearch, Logstash, Kibana)	ELK Stack	ELK Stack	ELK Stack
Monitor	Observability Platforms	Datadog, New Relic, Dynatrace	Datadog, New Relic, Dynatrace	Datadog, New Relic, Dynatrace	Datadog, New Relic, Dynatrace
Monitor	Incident Alerting	Alertmanager, PagerDuty, Opsgenie	Alertmanager, PagerDuty, Opsgenie	Alertmanager, PagerDuty, Opsgenie	Alertmanager, PagerDuty, Opsgenie

🔑 Key Observations by Language

Language	Build System	Testing	CI/CD	Special Notes
C++	CMake + Conan/vcpkg is the gold standard. Bazel for massive monorepos.	Google Test for unit tests. Valgrind & AddressSanitizer for memory debugging.	GitHub Actions & GitLab CI are most popular.	C++ DevOps is heavily focused on build speed (Ninja, ccache) and dependency management .
Python	Poetry is modern standard. Setuptools for legacy.	pytest is the undisputed king. Ruff is the new ultra-fast linter.	GitHub Actions is #1. Ansible & Fabric for automation.	Python shines in automation scripts and cloud SDKs (boto3, kubernetes client).
JavaScript	Vite is now #1 (98% usage). esbuild (88%) and Rollup (79%) are also huge. Webpack is declining (55%).	Jest is most popular. Vitest is gaining fast.	GitHub Actions & GitLab CI dominate. Vercel/Netlify for frontend hosting.	JavaScript DevOps is about fast builds and frontend deployment . TypeScript is now assumed.
AI / ML	Conda & Poetry for Python dependencies.	pytest for code. Great Expectations for data validation.	GitHub Actions for code CI. Kubeflow, Airflow, MLflow for ML pipelines.	MLOps is a superset of DevOps. Key additions: Model Serving (Triton, TF Serving), Experiment Tracking (MLflow, Weights & Biases), and Data Versioning (DVC, LakeFS).

🧠 AI's Role IN DevOps (Not Just FOR AI)

AI is now embedded **inside** DevOps tools themselves:

Tool	AI Feature
GitHub Copilot	Writes YAML pipelines, Dockerfiles, and Terraform code.
Amazon Q Developer	Expert in IaC (Terraform/CDK) and AWS.
Dynatrace / Datadog	AI-powered anomaly detection and root cause analysis.
Snyk AI	AI-assisted security vulnerability fixing.

Summary

Your Original List	DevOps Category	Tools (2025)
Git	Version Control	Git, GitHub, GitLab
C++, Python, JavaScript	Build + Test + Code Quality	CMake/Conan (C++), Poetry/pytest (Python), Vite/Jest (JS)
Web-based environment, IDE	Local Dev Environment	VS Code, Cursor, WebStorm, PyCharm
Library, packages setup	Package Managers	Conan/vcpkg, pip/Poetry, npm/yarn
Build system	Build Tools	CMake/Ninja, Vite/esbuild, Setuptools
Integration	CI/CD	GitHub Actions, GitLab CI, Jenkins
Cloud computing	IaC + Cloud	Terraform, Kubernetes, AWS/Azure/GCP
Testing	Testing Frameworks	Google Test, pytest, Jest
Debugging	Debugging + Tracing	GDB/Valgrind, pdb/Sentry, Jaeger/Zipkin
Design Management	Architecture (Not DevOps)	Handled by Software Architects
Consumer Satisfaction	Product Management (Not DevOps)	Handled by PMs & UX Researchers
Project Management	Planning & Tracking	Jira, Linear, Azure DevOps
Monitoring	Observability	Prometheus + Grafana, ELK, Datadog

This table gives you a **complete, language-specific roadmap** for mastering DevOps in 2025. Would you like me to go deeper into any specific category (e.g., how to set up a CI/CD pipeline for C++ with CMake + GitHub Actions)?